

TC2006B: Device Communications

Cables, Switches, VLANs, & Routers

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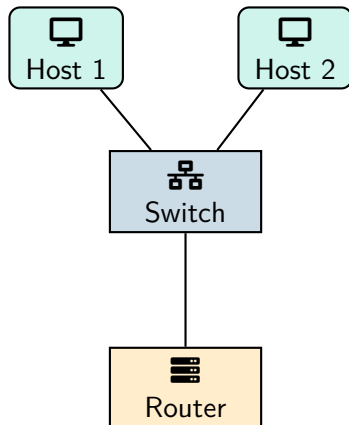
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Bridging the Gap: Mininet to Reality

What we did in Mininet:

- Created virtual topologies (nodes, links).
- Configured virtual switches (Open vSwitch).
- Assigned IP addresses to virtual hosts.
- Pinged between nodes.

The Goal of Today: Understand the real-world physical and logical equivalents of what Mininet simulates before we dive into Transport Layer services (TCP/UDP).



Layer 1: The Physical Layer (Cables)

The Physical Layer is responsible for moving raw bits (0s and 1s) over a physical medium. In Mininet, this is simulated by "virtual Ethernet" (veth) links.

Medium	Type	Typical Distance	Bandwidth
UTP (Cat6)	Copper (Electrical)	100m	1 Gbps - 10 Gbps
Coax	Copper (Electrical)	500m+	100 Mbps+
Fiber (Single Mode)	Glass (Light)	40km+	100 Gbps+
Fiber (Multi Mode)	Glass (Light)	300m - 500m	10 Gbps - 100 Gbps
Radio (Wi-Fi)	Wireless (RF)	100m	100 Mbps - 1 Gbps+

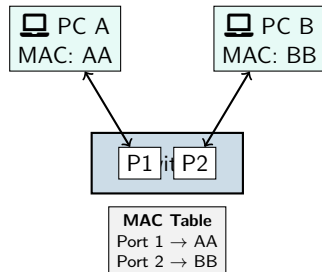
Key Concept

The physical layer doesn't understand IPs, MACs, or frames. It only understands voltage levels, light pulses, or radio frequencies representing binary data.

Layer 2: The Data Link Layer (Switches)

The Role of a Switch:

- Connects devices within the **same** network (Local Area Network - LAN).
- Operates using **MAC Addresses** (Hardware addresses).
- Breaks up **collision domains** (each port is its own collision domain).
- Forwards traffic intelligently based on a **MAC Address Table**.



MAC Learning

When a switch receives a frame, it records the *Source MAC* and the port it arrived on. It uses the *Destination MAC* to forward the frame out the correct port.

The Standard Ethernet Frame (Layer 2)

Switches forward **Frames**. The standard payload is up to 1500 bytes.

Dest. MAC (6B)	Src. MAC (6B)	EtherType (2B)	Payload (L3 Packet) 46 - 1500 Bytes	FCS (CRC) (4B)
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- **Dest/Src MAC:** 48-bit hardware addresses (e.g., 00:1A:2B:3C:4D:5E).
- **EtherType:** Tells the switch what is inside the payload (e.g., 0x0800 is IPv4).
- **FCS (Frame Check Sequence):** Used for error detection.

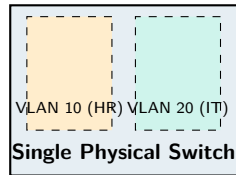
Virtual Local Area Networks (VLANs)

What is a VLAN?

At the simplest level, a VLAN is a way to **partition a single physical switch into multiple logical switches**.

Applications & Benefits:

- **Security:** VLANs do not forward traffic to members of other VLANs, preventing eavesdropping.
- **Flexibility:** Group users by function (e.g., HR, IT) regardless of their physical desk location.
- **Broadcast Control:** Limits the size of broadcast domains, improving network performance.



VLAN Access vs. Trunk Ports

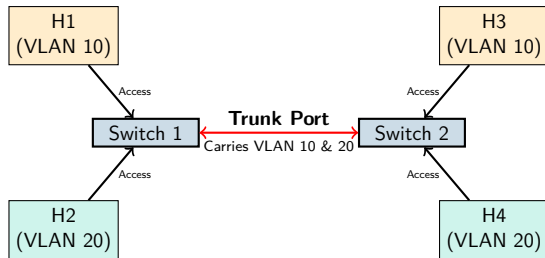
How do we span VLANs across multiple switches? (Like Open vSwitch in your labs).

Access Ports:

- Connected to end hosts (PCs, Servers).
- Assigned to **only one** VLAN.
- Traffic is untagged (normal Ethernet).

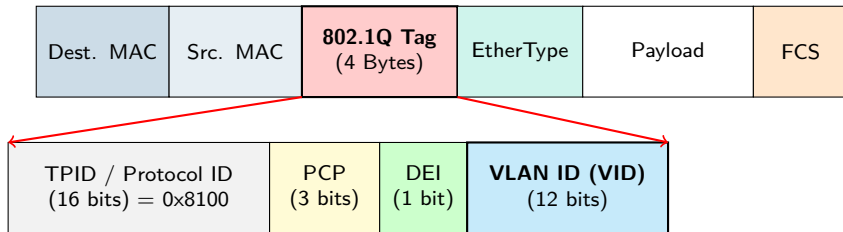
Trunk Ports:

- Connect switches to other switches or routers.
- Carry traffic for **multiple** VLANs simultaneously.
- Traffic is **tagged** to identify the VLAN.



VLAN Tagging (802.1Q)

To multiplex VLANs on a trunk port, the switch inserts a 4-byte **802.1Q Tag** into the original Ethernet frame.



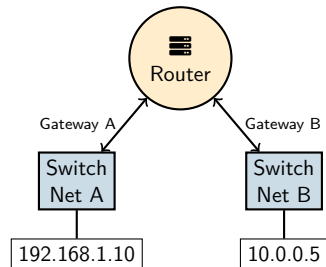
- **Protocol ID:** Identifies the frame as 802.1Q tagged (Always 0x8100).
- **VLAN ID (VID):** 12 bits allows for $2^{12} = 4096$ possible VLANs (0 - 4095).

Layer 3: The Network Layer (Routers)

If switches connect devices in the *same* network (or VLAN), how do devices in *different* networks communicate? **Routers**.

The Role of a Router:

- Connects **different** networks (Subnets).
- Operates using **Logical Addresses (IP Addresses)** (e.g., 192.168.1.10).
- Breaks up **Broadcast Domains** (Broadcasts don't cross routers).
- Uses a **Routing Table** to determine the best path to a destination.



Summary: Tying it all back to Mininet

When you build a topology in Mininet, you are scripting a full stack:

Layer	Hardware Reality	Addressing	Mininet Equivalent
Layer 3 (Network)	Routers route packets between subnets	IP Address	<code>h1.setIP('10.0.0.1/24')</code>
Layer 2 (Data Link)	Switches forward frames, VLANs partition	MAC Address	<code>net.addSwitch('s1')</code> (Open vSwitch)
Layer 1 (Physical)	Cables (Copper, Fiber) transmit signals	Bits	Links / veth pairs